WIRSCHING'S WEAK COVERING CONJECTURE: AN EXTENDED COMPUTATION, A CONDITIONAL BOUND, AND THE STRUCTURE OF WHAT RESISTS IT

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Abstract. Wirsching's 1998 Weak Covering Conjecture asks for a sub-exponential $K(l)$ such that a large enough set $R_{j-1,j} \subset \mathbb{Z}_3^*$, built from strictly decreasing exponent tuples, is forced to cover every unit residue modulo 3^l. We extend the exact computation of $j^*(l)$, the smallest such covering budget, from $l = 20$ to $l = 23$, and compare four growth models for $e(l) := j^*(l) - l\log_4 3$ against the extended table: a constant model is unconditionally impossible (we prove $e(l) \geq \frac{1}{4}\log_2 l - O(1)$) and disfavored at every statistic checked among the growing alternatives, which remain statistically indistinguishable from one another. We give the best bound known to us, $j^*(l) \leq \frac{9}{8}l + O(1)$, via a mean-payoff-game relaxation of the covering budget whose proof rests on one computationally verified, not generally proven, correspondence. We then examine the natural Fourier approach to a tighter bound: a uniform square-root-cancellation argument is capped below even this bound, and a direct computation at one accessible level finds the bulk of the relevant exponential sum's mass sitting outside any sparse exceptional set. Separately, the residues that actually fail to be covered sit at extremal cells of low local hit-density; a phase-randomization experiment at a different, more tractable pair of parameters points the same way, toward phase structure that magnitude alone does not capture. We prove a theorem on the uncovered residues themselves, for budgets $j \geq l$: the set left uncovered at a budget is contained in both twice and four times the set left uncovered at the previous budget, giving an explicit recursive upper bound on $j^*(l)$ that we prove exact, given one further combinatorial property verified at every level checked, and empirically exact everywhere else. We falsify the natural cost-1 form of a local repair rule this theorem might suggest, with explicit witness-level exhaustive counts, prove that every last-uncovered residue is $\equiv 1 \pmod 3$ whenever $j^*(l) \geq l + 1$ (every level to which the argument applies but one), prove a two-class containment law modulo 9 on the same residues, and report further exact identities, some verified but not proven, governing the last residues to be covered. Two precisely stated open questions remain, each narrower than the conjecture itself. The Weak Covering Conjecture remains open.

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1. Introduction

Wirsching's 1998 monograph [1] studies the density of predecessor sets under the $3n + 1$ map through an averaging construction on the 3-adic integers. A limiting Markov chain built from that construction has an invariant density, and Wirsching's own route to positive predecessor density passes through a covering question about a family of finite subsets of $\mathbb{Z}_3^*$, the 3-adic units, developed further in a later paper of his own that isolates positive predecessor density as a self-contained target and states, as open conjectures, the gaps still unbridged between that construction and a proof [2]. For integers $j, k \geq 0$, write

$$R_{j,k} := \left\{ \sum_{i=0}^{j} 2^{a_i} 3^i \;\middle|\; j + k \geq a_0 > a_1 > \cdots > a_j \geq 0 \right\} \subset \mathbb{Z}_3^*. \tag{1}$$

Each element of $R_{j,k}$ is built from a strictly decreasing sequence of $j+1$ "digit positions" chosen from $\{0,\ldots,j+k\}$: the i-th chosen position, counting from the top, contributes 2^{a_i} scaled by 3^i. Wirsching's Weak Covering Conjecture (Conjecture 3.9, Chapter V, [1]) asks for a sub-exponential $K(l)$, meaning $\lim_{l\to\infty} K(l)e^{-\gamma l} = 0$ for every constant $\gamma > 0$ (Wirsching's own definition, same chapter), such that $|R_{j-1,j}| \geq K(l)\cdot 3^l$ already forces $R_{j-1,j}$ to cover every unit residue modulo 3^l; the cardinality on the left is that of the tuple family itself, before reduction, which by Proposition 1's bijection equals $\binom{2j}{j}$, not the (a priori much smaller) size of its image modulo 3^l. Writing $j^*(l)$ for the smallest j such that $R_{j-1,j}$ covers $(\mathbb{Z}/3^l\mathbb{Z})^*$ exactly (a quantity that exists for every l, unconditionally; Proposition 15 below), the conjecture's forcing form and the growth rate of $j^*(l)$ coincide once coverage, once achieved, persists at every larger budget: this holds for $j \geq l$ (Theorem 9, Section 6), which covers every level in Table 1, but is not established below l, the regime the conjectured rate would eventually enter. With that proviso, the conjecture concerns the growth rate of

$$e(l) := j^*(l) - l\log_4 3, \tag{2}$$

the excess of the true covering budget over the leading-order rate a cardinality argument alone predicts ($|R_{j-1,j}| = \binom{2j}{j} \sim 4^j/\sqrt{\pi j}$, so matching $2\cdot 3^{l-1}$ residues needs $j \sim l\log_4 3$ at leading order). At the threshold, $K(l) \asymp |R_{j^*(l)-1,j^*(l)}|/3^l \asymp 4^{e(l)}/\sqrt{j^*(l)}$, a polynomial factor in l away from $4^{e(l)}$; since a polynomial factor never changes whether a sequence is sub-exponential, this makes $K(l)$ sub-exponential a consequence of the conjecture, unconditionally: the conjecture's truth would force $e(l)$ to grow slower than linearly, not just to grow. The converse, that $e(l) = o(l)$ would in turn establish the conjecture, needs coverage to persist at every budget past $j^*(l)$, which Theorem 9 only gives for $j \geq l$; below l, the regime a truly sub-linear $e(l)$ would eventually reach, the converse is open.

An earlier, unpublished manuscript of the author computed $j^*(l)$ exactly for $l = 1,\ldots,20$ by direct enumeration and reported the resulting table without a growth-rate analysis. This paper does three things. First, it extends that table to $l = 23$ and gives a proper statistical comparison of candidate growth models for $e(l)$ against the full range (Sections 2 and 3). Second, it gives the best upper bound on $j^*(l)$ we have found, via a reformulation of the covering budget as a mean-payoff game (Section 4). Third, and at greatest length, it reports a sustained attempt to close the gap to the conjectured rate directly, first by the natural Fourier approach, and, once that approach is shown to fail for a precise and checkable reason, by a direct study of the residues that actually resist coverage (Sections 5 and 6). That second attempt does not settle the conjecture. It produces a theorem on those residues, a falsification of the cost-1 form of a repair rule the theorem might suggest, and two precisely named open questions, each narrower than the conjecture itself and worth a dedicated attack in its own right.

2. The computation

Table 1 gives $j^*(l)$ and $e(l)$ for $l = 1,\ldots,23$. The values for $l \leq 20$ reproduce the earlier manuscript's table exactly; they were independently recomputed here by a from-scratch Rust reimplementation of the covering search (a bitset rotation dynamic program over residues modulo 3^l), validated against the original Python implementation at every $l \leq 20$ before being trusted for the extension. The values for $l = 21, 22, 23$ are new.

The single plateau in the table, $j^*(15) = j^*(16) = 20$, is not an error: $l = 15 \to 16$ is the only increment among the 22 steps from $l = 1$ to $l = 23$ at which the covering budget does not increase, and it is exactly the event Section 3's plateau-frequency test measures.

$l = 21$ fits in physical RAM once a memory-wasteful parallelization step in the original implementation was corrected. $l = 22$ required 500 GiB of swap, added to the machine specifically for this computation, and took approximately 10,749 s (~ 3 hours) with swap I/O as the dominant

l	$j^*(l)$	$e(l)$	l	$j^*(l)$	$e(l)$	l	$j^*(l)$	$e(l)$
1	1	0.208	9	13	5.868	17	21	7.528
2	4	2.415	10	15	7.075	18	22	7.735
3	6	3.623	11	16	7.283	19	23	7.943
4	7	3.830	12	17	7.490	20	24	8.150
5	9	5.038	13	18	7.698	21	25	8.358
6	10	5.245	14	19	7.905	22	26	8.565
7	11	5.453	15	20	8.113	23	27	8.773
8	12	5.660	16	20	7.320			

TABLE 1. $j^*(l)$ and $e(l) = j^*(l) - l\log_4 3$, $l = 1, \ldots, 23$.

cost, against 748 s for $l = 21$ without swap. $l = 23$ required approximately 263 GiB of state and 375,615 s ($\sim$104 hours); the successful run had no interruption, so a checkpoint/resume mechanism added after an earlier, unrelated policy-triggered kill of an interrupted attempt at the same level was never exercised on it. Three attempts at $l = 24$, made over the course of this project on the same machine, each failed before completion (two to memory-policy kills, one to an apparent out-of-memory condition with no surviving kernel log confirming it directly); $l = 24$ is not attempted further, and the table stops at $l = 23$.

3. THE GROWTH OF $e(l)$

We compare four candidate functional forms for $e(l)$'s growth against the tail $l = 10, \ldots, 23$ ($n = 14$; the same range choice as the earlier manuscript, now with three more points), each fit by ordinary least squares: $e(l) = c$ (constant), $e(l) = a + b\ln l$ (logarithmic), $e(l) = a + b\sqrt{l}$ (square-root), and $e(l) = a + bl$ (slow-linear). Table 2 gives the fit statistics.

model	k	ΔAIC	ΔBIC	LOOCV RMSE
constant	1	16.091	15.452	0.5200
logarithmic	2	1.512	1.512	0.2991
square-root	2	0.722	0.722	0.2909
slow-linear	2	0.000	0.000	0.2837

TABLE 2. Model comparison on $e(l)$, $l = 10, \ldots, 23$. ΔAIC/BIC relative to the best-fitting model (slow-linear).

A constant model is disfavored sharply and consistently: ΔAIC = 16.09, ΔBIC = 15.45, both growing at every l added since $l = 21$. The three growth models remain statistically indistinguishable from one another by the conventional ΔAIC > 2 rule of thumb, with slow-linear taking a narrow lead on both AIC and leave-one-out cross-validation; the three candidate regressors ($\log l$, $\sqrt{l}$, l) are themselves highly correlated over this range (pairwise correlation ≥ 0.993), so this ranking says more about how thin the margin currently is than about which functional form is correct. The stakes are not merely descriptive: by the identity noted after (2), a linear $e(l)$ would make $K(l)$ exponential, falsifying the conjecture outright rather than leaving it merely unproven. Fourteen highly correlated points cannot distinguish that possibility from logarithmic or square-root growth, so this comparison does not settle the question either way; it only says which growth rates remain live.

None of this machinery is actually needed to rule out a constant $e(l)$: since $\log_4 3 < 1$, rounding a constant-plus-linear quantity $l\log_4 3 + c$ to the nearest integer can only change by 0 or 1 from one l to the next, yet $j^*(1) = 1$ and $j^*(2) = 4$, an increment of 3. A constant-$e(l)$ source is incompatible with the table's first two entries alone, with no statistical test required.

A direct combinatorial test tells the same story from a different angle, restricted to where its own null hypothesis is not already dead: the constant-rounding model behind it predicts every increment in $\{0,1\}$, which the full table already refutes (the increments of 2 and 3 noted above), so a plateau count over the full range is not a meaningful test of that model. Restricted to the $l = 10, \dots, 22$ tail the model comparison above uses (13 increments over the same 14 points, all actually in $\{0,1\}$, consistent with the model's own prediction), exactly one is zero (the $l = 15 \to 16$ plateau noted above); the earlier manuscript's own table, $l \le 20$, showed 3.9 plateaus expected in 19 increments under the same model, and testing the observed count against $\mathrm{Bin}(13, 3.9/19)$, one-sided, gives $p \approx 0.22$. $e(l)$ is an exact, deterministic sequence, so this carries no sampling interpretation; it is best read descriptively, as "how well does a constant-$e(l)$ source fit the observed tail increment pattern," and it is not independent corroboration of the model comparison above: both look at the same 14-point tail.

A constant $e(l)$ is disfavored descriptively by every measure above. It is also, unconditionally, impossible.

Proposition 1 (Unboundedness of $e(l)$)**.** *$e(l) \ge \frac{1}{4}\log_2 l - O(1)$.*

Proof. Distinct exponent tuples give distinct sums: since the a_i are distinct, a_{j-1} is the unique minimal one, so $V = \sum_i 2^{a_i}3^i$ has 2-adic valuation exactly a_{j-1} (every other term is divisible by a strictly higher power of 2), recovering a_{j-1} from V directly. The full term $2^{a_{j-1}}3^{j-1}$ is then subtracted off, and the process iterates on the remaining $j-1$ terms. So $R_{j-1,j}$, in bijection with j-subsets of the $2j$-element set $\{0, \dots, 2j-1\}$, has exact cardinality $\binom{2j}{j}$. Covering $(\mathbb{Z}/3^l\mathbb{Z})^*$, of size $2 \cdot 3^{l-1}$, requires the domain to be at least as large as the codomain: $\binom{2j}{j} \ge 2 \cdot 3^{l-1}$ is necessary at $j = j^*(l)$ in particular, since $R_{j^*(l)-1,j^*(l)}$ covers by definition of $j^*(l)$. The explicit bound $\binom{2n}{n} \le 4^n/\sqrt{3n+1}$, valid for every $n \ge 1$, turns this into $4^j \ge 2 \cdot 3^{l-1}\sqrt{3j+1}$, hence

$$j \ge l \log_4 3 - \log_4 3 + \log_4 2 + \tfrac{1}{2}\log_4(3j+1).$$

Dropping the last, non-negative term already gives $j \ge l \log_4 3 - O(1)$, so $3j + 1 = \Omega(l)$, hence $\log_4(3j+1) \ge \log_4 l - O(1) = \frac{1}{2}\log_2 l - O(1)$. Feeding this back into the displayed inequality gives the claimed bound. $\square$

4. A mean-payoff-game upper bound

The full-precision game. Fix $l \ge 1$. A *full-precision play* from a unit z_0 modulo 3^l is a sequence of l moves: at step $i = 0, \dots, l-1$, from a unit z_i modulo 3^{l-i}, a move of cost $d_i \ge 0$ is *legal* when $2^{d_i}z_i \equiv 1 \pmod 3$, and produces $z_{i+1} := T_{d_i}(z_i) := (2^{d_i+1}z_i - 2)/3$, which is then required to be a unit modulo 3^{l-i-1} (automatic at $i = l-1$, where $3^{l-i-1} = 1$). Write $J := \sum_{i=0}^{l-1} d_i$ for the total cost.

Lemma 2 (Grounding)**.** *For a legal full-precision play from z_0 with costs $d_0, \dots, d_{l-1}$ and total cost J, define $a_{l-1} := 0$ and $a_{i-1} := a_i + d_i + 1$ for $i = l-1, \dots, 1$ (so $a_0 = J - d_0 + l - 1$). Then*

$$2^{J+l-1}z_0 \equiv \sum_{i=0}^{l-1} 2^{a_i}3^i \pmod{3^l}.$$

The right side is, by construction, an element of $U(l, a_0)$ (Section 7) with bottom exponent $a_{l-1} = 0$.

Proof. By induction on the play length n ($n = 1, \dots, l$), stated for a play of length n starting at a unit z modulo 3^n (matching the lemma's notation at $n = l$, $z = z_0$). Write $J_n := \sum_{i=0}^{n-1} d_i$ and $a_{n-1} := 0$, $a_{i-1} := a_i + d_i + 1$ for $i = n-1, \dots, 1$, as above.

Base case $n = 1$: the single move has $2^{d_0}z \equiv 1 \pmod 3$, which is exactly $2^{a_0+d_0}z \equiv 2^{a_0}3^0 \pmod 3$ since $a_0 = 0$ when $n = 1$.

Inductive step: suppose the claim holds at length $n-1$ for the sub-play with costs $d_1, \ldots, d_{n-1}$ starting at $z' := T_{d_0}(z)$, a unit modulo 3^{n-1}; writing $a'_i := a_{i+1}$ for this sub-play, $a'_{n-2} = a_{n-1} = 0$ matches its base case, and its recursion $a'_{i-1} = a'_i + d_{i+1} + 1$ matches $a_i = a_{i+1} + d_{i+1} + 1$ (the original recursion at index $i+1$), so the induction hypothesis reads

$$2^{a_1+d_1} z' \equiv \sum_{i=0}^{n-2} 2^{a_{i+1}} 3^i \pmod{3^{n-1}}.$$

By definition of T_{d_0}, $2^{d_0+1} z \equiv 3z' + 2 \pmod{3^n}$. Multiplying by the unit $2^{a_1+d_1}$ gives

$$2^{a_1+d_1+d_0+1} z \equiv 3 \cdot 2^{a_1+d_1} z' + 2^{a_1+d_1+1} \pmod{3^n}.$$

Since $a_0 = a_1 + d_1 + 1$ (the recursion at $i = 1$), the left side is $2^{a_0+d_0} z$ and the last term is 2^{a_0}. Multiplying the induction hypothesis by 3 turns it, modulo 3^n, into $3 \cdot 2^{a_1+d_1} z' \equiv \sum_{i=1}^{n-1} 2^{a_i} 3^i$, so

$$2^{a_0+d_0} z \equiv 2^{a_0} + \sum_{i=1}^{n-1} 2^{a_i} 3^i = \sum_{i=0}^{n-1} 2^{a_i} 3^i \pmod{3^n},$$

the claim at length n, using $a_0 + d_0 = J_n + n - 1$ (telescoping the recursion). Verified directly against every legal play for $l = 1, \ldots, 4$ as a check, with no exception, before being trusted. □

Empirical Result 3 (The game's worst case is $j^*(l)$).

$$j^*(l) = \max_{z_0} \min\{J : z_0 \text{ has a legal full-precision play of total cost } J\},$$

the maximum over units z_0 modulo 3^l. Verified against the known exact table of $j^*(l)$ for $l = 1, \ldots, 12$, with no discrepancy. Lemma 2 does not by itself establish this: the twist 2^{J+l-1} in the lemma depends on z_0 through its own optimal J, so it is not a single, z_0-independent relabeling of the unit group, and we do not have a general proof of the displayed identity.

The window-k relaxation. A move of cost $d \geq 0$ at z is *legal* when $2^d z \equiv 1 \pmod 3$, as above, and *safe* at window k when, for $m := 3^k$, every one of the three residues $z, z+m, z+2m$ modulo $3m$ maps under T_d back into a unit modulo m. A minimizing player, seeing only $z \bmod 3^k$, chooses a legal, safe d; an adversary then reveals the hidden next 3-adic digit $\epsilon \in \{0, 1, 2\}$, forcing the true state to $T_d(z + 3^k \epsilon) \bmod 3^k$. Both requirements matter: legality alone can send a unit to 0 (take $z = 1$, $d = 0$: $T_0(1) = 0$), and the safety filter is exactly what keeps the walk inside $\mathbb{Z}_3^*$ regardless of which digit the adversary supplies. Legality depends on d only through $d \bmod 2$ (since 2 has order 2 modulo 3), and 2 is a primitive root modulo 3^{k+1} for every $k \geq 0$ (order $2 \cdot 3^k$, by an elementary lifting-the-exponent induction), so $2^{d+1} \bmod 3^{k+1}$, and hence $T_d(z) \bmod 3^k$ and the safety check itself, depend on d only through $d \bmod (2 \cdot 3^k)$: successor states repeat with this period as d grows, and every representative beyond the smallest in its residue class has strictly larger cost for the same effect, so is never used by an optimal policy. The action set at each state is therefore, without loss, the subset of $\{0, \ldots, 2 \cdot 3^k - 1\}$ that is legal and safe there, which is finite. That this subset is nonempty at every unit state has been verified computationally at every $k = 3, \ldots, 14$ used below, not proven in general here.

This restricted-visibility game is, by construction, a two-player *mean-payoff game* on the $2 \cdot 3^{k-1}$ unit states modulo 3^k, with the now finite legal, safe moves as the minimizer's edges, the adversary's three choices of ϵ as the maximizer's edges, and payoff the long-run average of the move costs d. The classical theory of such games, initiated by Ehrenfeucht and Mycielski [5], guarantees a value and an optimal positional policy in general; Theorem 4 below needs none of that generality, only the twelve specific policies actually exhibited and directly checked in Table 3.

A window-k policy, run against the actual digit sequence, gives a legal full-precision play at any z modulo 3^l: at step i, the true remaining state is z_i modulo 3^{l-i}, so the policy sees the true $z_i \bmod 3^k$ directly whenever $l - i \geq k$, i.e. for every step but the last $k-1$. Once $l - i < k$, whether because $l < k$ globally or because a play with $l \geq k$ has reached its own final $k-1$ steps, fix any lift of the true state to a residue modulo 3^k and run the policy on the lift, which is exactly the situation safety was built for, with the play's own remaining digits standing in for the hidden ones, so the chosen move stays legal against the true, lower-precision state at every step (this regime is part of what the policy-against-the-covering-DP check at $k = 4, 7$ against l up to 14 already exercised, with zero illegal or unsafe moves; every play there passes through this same final-$k-1$-steps regime regardless of how large l is).

Theorem 4 (Window relaxation bound, conditional on Empirical Result 3)**.** *For each $k = 3, \ldots, 14$, the certified policy underlying Table 3 (a legal, safe move at every unit state modulo 3^k, average cost ρ_k) gives an explicit rational C_k with $j^*(l) \leq \rho_k \cdot l + C_k$ for all l.*

Proof. Fix k in range. The policy σ_k, the value ρ_k, and a potential $h_k : (\mathbb{Z}/3^k\mathbb{Z})^* \to \mathbb{Q}$ are computed by a nested Howard strategy-improvement solver and checked, state by state and digit by digit over the finite action set established above, against

$$d_{\sigma_k}(s) + h_k(s'(\epsilon)) \leq \rho_k + h_k(s)$$

at every state s and every adversary digit ϵ, where $s'(\epsilon)$ is the successor under σ_k's chosen move; the matching adversary lower bound extracted from the same run certifies ρ_k exactly. Summing the inequality along any l-step play $s_0, \ldots, s_l$ telescopes to

$$\sum_{i=0}^{l-1} d_{\sigma_k}(s_i) \leq l\rho_k + h_k(s_0) - h_k(s_l) \leq l\rho_k + \big(\max h_k - \min h_k\big) =: l\rho_k + C_k,$$

regardless of which digits the adversary actually supplies, since the potential inequality holds for every ϵ at every step. So σ_k, run as above, gives every unit z modulo 3^l a legal full-precision play of total cost at most $l\rho_k + C_k$; the minimum cost over all legal plays from z is therefore also at most $l\rho_k + C_k$. Taking the worst-case z and applying Empirical Result 3 gives $j^*(l) \leq \rho_k l + C_k$. $\square$

ρ_k and C_k are computed exactly, as rationals, by a nested Howard strategy-improvement solver, self-certified at every k via a matching adversary lower bound extracted from the same run (upper bound equals lower bound, so the computed rational equals ρ_k exactly), and cross-validated by an independent value-iteration solver and, at $k = 4$ and $k = 7$, by running the resulting policy against the actual finite covering DP for every unit residue modulo 3^l, l up to 14, with zero illegal or unsafe moves encountered. Table 3 gives the values computed for $k = 3, \ldots, 14$.

k	3	4	5	6	7	8	9	10	11	12	13	14
ρ_k	2	$\frac{5}{3}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{7}{5}$	$\frac{25}{19}$	$\frac{5}{4}$	$\frac{11}{9}$	$\frac{6}{5}$	$\frac{7}{6}$	$\frac{119}{104}$	$\frac{9}{8}$
C_k	5	$\frac{19}{3}$	$\frac{15}{2}$	9	10	$\frac{207}{19}$	$\frac{47}{4}$	$\frac{115}{9}$	$\frac{69}{5}$	$\frac{44}{3}$	$\frac{1621}{104}$	$\frac{33}{2}$

TABLE 3. The window-relaxation value ρ_k and constant C_k, $k = 3, \ldots, 14$.

At $k = 14$, $\rho_{14} = 9/8 = 1.125$ and $C_{14} = 33/2$, giving:

Corollary 5 (Conditional on Empirical Result 3)**.** *$j^*(l) \leq \frac{9}{8}l + O(1)$.*

This is, to our knowledge, the best bound on $j^*(l)$'s growth rate currently available, conditional on Empirical Result 3. ρ_k is observed to be non-increasing over the computed range $k = 3, \ldots, 14$ (Table 3); we have not verified this in general, and Corollary 5 uses only the single value $k = 14$. The general technique, representing a covering-type combinatorial problem as a mean-payoff game and computing exact rational relaxation bounds this way, is not new to this application: beyond Ehrenfeucht and Mycielski's foundational theorem [5] itself, a close and recent methodological precedent proves rationality and computability of the covering radius of a primitive sofic shift by essentially the same construction, applied to a different object [6]. We cite both as precedent for the framing; what is new here is its specific application to $j^*(l)$.

Remark 6. Fitting $\rho_k = L + A/k$ by ordinary least squares to $k = 10, \dots, 14$ gives $L \approx 0.876$, still well above the conjectured true rate $\log_4 3 = 0.7925\dots$; the fit is sensitive to the range used (all of $k = 3, \dots, 14$ gives $L \approx 0.904$) and is not evidence of any specific limit. Only the direction $j^*(l) \le \rho_k l + C_k$, proven above, is established. The reverse inequality, $\inf_k \rho_k \le \limsup_l j^*(l)/l$, which would be needed to conclude anything about where ρ_k is heading, is not established, and there is evidence in the course of this work that the window relaxation may carry an irreducible gap above the true rate.

5. The exponential sum, and why it resists a direct attack

Corollary 5's bound, $1.125\,l$, is well above the conjectured rate $\log_4(3)\,l \approx 0.7925\,l$. The natural route to closing that gap is Fourier-analytic. Here $e(x) := e^{2\pi i x}$ denotes the standard additive character, unrelated to the sequence $e(l)$ of (2). By orthogonality on $\mathbb{Z}/3^l\mathbb{Z}$, the number of tuples landing in residue class z is

$$N(z) = \frac{1}{3^l}\sum_{t=0}^{3^l-1} S(t)\, e\!\left(-\frac{tz}{3^l}\right), \qquad S(t) := \sum_{j+k\ge a_0>\cdots>a_j\ge 0} e\!\left(\frac{t}{3^l}\sum_{i=0}^{j} 2^{a_i}3^i\right), \tag{3}$$

and, since $S(0) = |R_{j,k}| =: T$, the triangle inequality gives $N(z) > 0$ for every z as soon as $\sum_{t\ne 0}|S(t)| < T$; a uniform bound $|S(t)| = o(T/3^l)$ at every nonzero t would suffice, at a rate controlled directly by the exponent in the bound. This approach does not work, for three separate and mutually reinforcing reasons.

5.1. A hard ceiling on uniform cancellation.

Proposition 7. *Any proof of covering via a uniform bound $|S(t)| \le C\sqrt{T}$ for a fixed constant C (square-root cancellation), applied via the triangle inequality across all $3^l - 1$ nonzero frequencies, requires $j \ge 2\log_4(3)\,l - O(1) \approx 1.585\,l - O(1)$.*

Proof. Write $T := |R_{j-1,j}| = \binom{2j}{j}$. Positivity of $N(z)$ for every z via $\sum_{t\ne 0}|S(t)| < T$ combined with the uniform estimate $|S(t)| \le C\sqrt{T}$ over the $3^l - 1$ nonzero frequencies forces $(3^l-1)C\sqrt{T} < T$, so $T > ((3^l-1)C)^2$, i.e. $4^j/\sqrt{3j+1} \ge T > ((3^l-1)C)^2$ (using the explicit bound from Proposition 1's proof), which gives $j \ge 2\log_4(3)\,l - O(1)$. ☐

Proposition 7's threshold is *worse* than Corollary 5's bound: any approach that only delivers generic square-root cancellation across every frequency cannot even match what is already conditionally known by other means, let alone improve on it. This rules out the naive version of the Fourier approach cleanly, and directs attention to whether a finer, frequency-dependent treatment (a major/minor-arc split, in the language of the circle method) can do better.

5.2. A computation: the exceptional mass does not stay small.

Numerically, $|S(t)|$ is far from uniform: the frequencies of largest magnitude sit close to dyadic rationals of small denominator, $t/3^l \approx p/2^r$ for small r. The natural repair for Proposition 7's ceiling is to excise these major-arc resonances and control the remainder by an averaged bound; whether that remainder stays sparse enough at the scale that matters is a question this section answers computationally, for the family $R_{j-1,j}$ at $l = 18$, $j = 16$ (the first-order cardinality threshold, $T := \binom{32}{16}$), six budgets below the actual covering threshold $j^*(18) = 22$: coverage necessarily fails here already, and this level sits below the $j \ge l$ range Sections 6–7 otherwise work in), with a negative result. Write $\|S\|_1 := \sum_{\gcd(t,3)=1}|S(t)|/T$ for the normalized L^1 mass over frequencies coprime to 3; here $\|S\|_1 = 5226.01$. A fixed threshold makes the visible spikes look sparse, but of the 8,014 frequencies with $|S(t)|/T$ exceeding 0.001, their combined contribution to $\|S\|_1$ is under 132, so over 97% of the mass at this single computed level lives in an exponentially numerous population of near-average-magnitude coefficients, not in a sparse exceptional set. This is evidence against a sparse-exceptional-set repair at accessible l; it says nothing about every l.

The same clustering near dyadic rationals connects the problem to a 2011 discussion of Terence Tao's, applying Littlewood–Offord and Riesz-product methods to an object built from the same $2^a 3^b$-type exponential sums, in the closely related setting of a hypothetical non-trivial Collatz cycle [7]; his construction is structurally close to a boundary slice of $R_{j,k}$ with the bottom exponent pinned at 0. His own conclusion, in his own words, is that "once one uses the transcendence theory bound, ... Littlewood-Offord techniques are no longer of much use," and that he knows "of no way to get a nontrivial separation property between powers of 2 and powers of 3 other than via transcendence theory methods."

5.3. **A direct test of the residues that actually fail.** The most direct test examines the residues that actually fail to be covered, one budget step before covering succeeds, i.e. $H(l, j^*(l)-1)$ (Section 6). For $l = 10, \ldots, 15$ this set has size $2, 1, 3, 1, 3, 1$ respectively, tiny against a global occupancy mean $\lambda = |R_{j^*(l)-2,j^*(l)-1}|/(2 \cdot 3^{l-1})$ that is itself in the thousands over the same levels (from 1019 at $l = 10$ to 3695 at $l = 15$). Write the *depth-c cell* of a residue z for its class modulo 3^c, and its *depth-c local intensity* $\lambda_c(z)$ for the total hit count within that class divided by 3^{l-c}, the class's size (defined for $c < l$; at $c = l$ the class is $\{z\}$ itself, and $\lambda_l(z) = 0$ identically for any holdout, a degenerate case excluded below). A *non-homogeneous Poisson model* fit to that intensity assigns z hole probability $e^{-\lambda_c(z)}$. Computed directly from the exact histogram for $l = 10, \ldots, 15$, every one of the eleven holdouts across these six levels has λ_c well below the global mean at every depth $c = 8, 9, 10$ with $c < l$ (at $l = 10$, only $c = 8, 9$; λ_c from 2.0 to 108.4, hole probability from e^{-2} to e^{-108}): holdouts sit systematically in low-intensity cells, not typical ones, though how far below the mean varies substantially by depth and level rather than clustering around one exponent. A separate diagnostic at a smaller, more tractable pair, $(l, m) = (14, 16)$ (mean hit count 188.5, a budget below the covering threshold $j^*(14) = 19$), holds every $|S(t)|$ fixed and randomizes the phases of conjugate-symmetric pairs: the actual primitive-frequency max/RMS ratio is 15.79, while eight independent phase scrambles (fixed seed) give ratios only in $[5.11, 5.24]$, collapsing to what a generic-phase model predicts once the phases are scrambled away. At the levels checked, both the raw holdout rarity and the phase-scramble gap point the same way: coverage failure is decided by phase structure that magnitude-only methods (uniform bounds, averages, low moments of $S(t)$) do not see.

Proposition 8. *$|S(3^{l-1})|/\binom{2m}{m} \to 1/\sqrt{3}$ as $m \to \infty$, for the family $R_{m-1,m}$, for every fixed $l \geq 1$.*

Proof sketch. $\sum_i 2^{a_i} 3^i \equiv 2^{a_0} \pmod 3$, so $S(3^{l-1})/\binom{2m}{m}$ depends only on the limiting distribution of the top exponent a_0's parity as $m \to \infty$, which converges to the mixture $\frac{1}{3}\omega + \frac{2}{3}\omega^2$ of the two nontrivial cube roots of unity $\omega = e^{2\pi i/3}$ (the limiting gap $2m-1-a_0$ of the maximum of a random m-subset of $\{0, \ldots, 2m-1\}$ is geometric with parameter $1/2$, giving $\Pr[a_0 \text{ even}] \to 1/3$). This mixture equals $-\frac{1}{2} - \frac{i\sqrt{3}}{6}$, of modulus $1/\sqrt{3}$. □

Uniform decay across every nonzero frequency, the premise a naive statement of the Fourier approach might suggest, is false at this explicit family of frequencies: $t/3^l = 1/3$ exactly, so l drops out of $S(3^{l-1})$ entirely (only V mod 3 matters), and the limit above holds at every scale m, including along the covering-relevant diagonal $m \sim l \log_4(3)$.

6. A theorem on the residues that resist coverage

Write $H(l, j)$ for the set of units modulo 3^l *not* covered by $R_{j-1,j}$, so that $j \geq j^*(l)$ exactly when $H(l, j) = \emptyset$. Every result in this section needs $j \geq l$; at every computed level $j^*(l) \geq l$, with equality only at $l = 1$ (Table 1), but the conjectured rate $\log_4 3 < 1$ would eventually put $j^*(l)$ below l, outside this section's reach. Within this range, emptiness persists: Theorem 9 below gives

$H(l,j)=\emptyset \Rightarrow H(l,j+1) \subseteq 2\emptyset \cap 4\emptyset = \emptyset$, so "$H(l,j)=\emptyset$ exactly when $j \geq j^*(l)$" is consistent for every $j \geq l$. We make no claim about $j < l$.

Theorem 9 (Holdout doubling inclusion)**.** *For $j \geq l$, $H(l,j+1) \subseteq 2H(l,j) \cap 4H(l,j)$.*

Proof. $R_{j-1,j}$ is the set of values $r=\sum_{i=0}^{j-1} 2^{b_i}3^i$ over j-subsets $\{b_0 > \cdots > b_{j-1}\} \subseteq \{0,\ldots,2j-1\}$, and $R_{j,j+1}$ the same with $(j+1)$-subsets of $\{0,\ldots,2j+1\}$. Fix $r \in R_{j-1,j}$ with exponent set S, and $s \in \{1,2\}$. Shifting every exponent up by s gives $S+s \subseteq \{s,\ldots,2j-1+s\} \subseteq \{1,\ldots,2j+1\}$, a j-subset whose induced value is $2^s r$ (a uniform shift of exponents preserves relative rank, so each term's power of 3 is unchanged). Since $S+s$ contains no 0, adjoining 0 as the new smallest exponent gives a $(j+1)$-subset of $\{0,\ldots,2j+1\}$: it becomes the new rank-j term, contributing $2^0\cdot 3^j = 3^j$. So this witness represents $2^s r + 3^j$, and $3^j \equiv 0 \pmod{3^l}$ once $j \geq l$. Hence $2R_{j-1,j} \cup 4R_{j-1,j} \subseteq R_{j,j+1} \pmod{3^l}$ for $j \geq l$. So if $z \in H(l,j+1)$, i.e. z is not represented at budget $j+1$, then z lies in neither $2R_{j-1,j}$ nor $4R_{j-1,j}$: both $z/2$ and $z/4$ are unrepresented at budget j, i.e. $z \in 2H(l,j) \cap 4H(l,j)$. ☐

Corollary 10 (Chain contraction)**.** *For $j \geq l$: if $x, 2x, \ldots, 2^{t-1}x \in H(l,j+1)$ are pairwise distinct (all reduced modulo 3^l, $t \geq 1$), then $x/4, x/2, x, \ldots, 2^{t-2}x \in H(l,j)$ are pairwise distinct, a chain of length $t+1$.*

Proof. Apply Theorem 9 to each of the t elements: $2^i x \in H(l,j+1)$ gives $2^{i-1}x, 2^{i-2}x \in H(l,j)$ for $i=0,\ldots,t-1$. The union of these pairs over all i is exactly $\{x/4, x/2, x, \ldots, 2^{t-2}x\}$, $t+1$ elements indexed by $t+1$ consecutive powers of 2. Since 2 has order $2\cdot 3^{l-1}$ modulo 3^l (an elementary lifting-the-exponent fact, used already in Section 4), any $2\cdot 3^{l-1}$ consecutive powers of 2 times x exhaust every unit modulo 3^l, and any shorter run is pairwise distinct. The hypothesis that $x, 2x, \ldots, 2^{t-1}x$ are pairwise distinct already forces $t \leq 2\cdot 3^{l-1}$, so it remains to rule out equality. If $t = 2\cdot 3^{l-1}$, those t elements would be every unit modulo 3^l, including every element of the (nonempty) image of $R_{j,j+1}$; but $H(l,j+1)$, the complement of that image, contains no such element. Hence $t < 2\cdot 3^{l-1}$, and the $t+1$ output elements, a shorter run of consecutive powers, are pairwise distinct. ☐

Corollary 11 (Run-length bootstrap)**.** *Writing* $\operatorname{maxrun}(H)$ *for the greatest t such that H contains t pairwise distinct elements of the form $x, 2x, \ldots, 2^{t-1}x$ (all reduced modulo 3^l), with* $\operatorname{maxrun}(\emptyset) := 0$*, $j^*(l) \leq j + \operatorname{maxrun}(H(l,j))$ for every $j \geq l$.*

Proof. Let $r := \operatorname{maxrun}(H(l,j))$ and suppose, for contradiction, $H(l,j+r) \neq \emptyset$; pick $z \in H(l,j+r)$, a chain of length 1 ($t=1$) in the sense of Corollary 10. Applying Corollary 10 once at each of the r budgets $j+r-1, j+r-2, \ldots, j$ (all $\geq l$, since $j \geq l$), a chain of length t at budget $j'+1$ becomes a chain of length $t+1$ at budget j' each time, so after r applications the length-1 chain at $j+r$ becomes a chain of length $r+1$ in $H(l,j)$, contradicting $\operatorname{maxrun}(H(l,j)) = r$. So $H(l,j+r)=\emptyset$, i.e. $j^*(l) \leq j+r = j+\operatorname{maxrun}(H(l,j))$. ☐

Empirical Result 12. Corollary 11 is empirically exact, $j^*(l) = j + \operatorname{maxrun}(H(l,j))$, at every (l,j) with $l+1 \leq j \leq j^*(l)$ we were able to compute, $l = 5,\ldots,20$ at every such budget and l up to 22 at the single budget $j = l+1$, including a case with $|H(l,j)| > 3\times 10^6$. (Past $j^*(l)$, $H(l,j)=\emptyset$ and $\operatorname{maxrun}(\emptyset)=0$, so the displayed equality no longer holds there; the claim is only about budgets up to and including $j^*(l)$.) The lower endpoint $j=l+1$ is not merely where computation started: equality fails at $j=l$ (e.g. $\operatorname{maxrun}(H(6,6)) = 5$, giving $j + \operatorname{maxrun} = 11 \neq 10 = j^*(6)$; $\operatorname{maxrun}(H(7,7)) = 5$, giving $12 \neq 11 = j^*(7)$), consistent with Corollary 10 and Corollary 11 both requiring $j \geq l$. The sub-range $l+2 \leq j \leq j^*(l)$ is proven, not merely empirical, at $l = 3,\ldots,13$: Proposition 22 below. The single boundary budget $j = l+1$ remains empirical only, tied to Proposition 20's own open converse.

The converse direction, that a doubling chain of the observed maximal length is not merely sufficient but necessary for survival to that budget, is not proven, and is false as an unrestricted statement: $1547 \in 2H(7,8) \cap 4H(7,8)$, yet $1547 \notin H(7,9)$ (both of 1547's two budget-9 witnesses span the full exponent range, the one case Theorem 9's argument does not directly control). We return to a restricted, still-open form of this converse in Section 7.

A natural strengthening of Theorem 9, that every individual witness admits a bounded-cost local repair, is also false in its most optimistic form.

Call the *Durfee depth* of a witness $\{a_0 > \cdots > a_{j-1}\} \in R_{j-1,j}$ at budget j the count $\#\{i : a_i \geq j\}$, the number of chosen exponents at or above the budget itself.

Empirical Result 13 (Failure of the cost-1 local repair rule). At the budget transition $(l, j) = (6, 10) \to (7, 11)$, call a unit modulo 3^7 a *child* and its reduction modulo 3^6 its *parent*; 100 of the 1458 children (all units modulo 3^7) have minimum achievable Durfee depth exceeding their parent's minimum depth by more than one, up to three; at $(7, 11) \to (8, 12)$, 332 of 4374 children (all units modulo 3^8) do, likewise up to three. A correspondingly exhaustive check, measuring witness repair cost as the minimum number of exponents that must change between a child's cheapest witness and its cheapest parent witness (matched by the natural containment of exponent sets), finds distances up to 7 at both transitions, against a distance of 1 for the overwhelming majority. This is evidence against a repair cost of exactly 1 at these two transitions; it does not establish that no bounded repair cost exists.

This falsifies, in its originally proposed cost-1 form, a natural conjecture this project had earlier entertained (that coverage propagates via a uniform, cost-1 local repair rule budget by budget); its valid surviving content is Theorem 9 and its corollaries above.

7. Further structure of the last holdouts

The residues in $H(l, j^*(l) - 1)$, the last to be covered, satisfy several further exact laws. For $W \geq l - 1$, write

$$U(l, W) := \left\{ \sum_{i=0}^{l-1} 2^{\beta_i} 3^i \bmod 3^l \;\middle|\; W \geq \beta_0 > \cdots > \beta_{l-1} \geq 0 \right\},$$

an l-term analogue of $R_{j-1,j}$ built from a fixed term count and a growing exponent ceiling W.

Lemma 14 (Width scaling identity)**.** *For $j \geq l \geq 1$,*

$$R_{j-1,j} \equiv 2^{j-l}\, U(l, j + l - 1) \pmod{3^l}. \tag{4}$$

Proof. Since $3^i \equiv 0 \pmod{3^l}$ for $i \geq l$, the residue of $r = \sum_{i=0}^{j-1} 2^{b_i} 3^i \in R_{j-1,j}$ modulo 3^l depends only on its top l exponents $b_0 > \cdots > b_{l-1}$. A decreasing l-tuple $b_0 > \cdots > b_{l-1} \geq 0$ with $b_0 \leq 2j - 1$ extends to a full admissible j-tuple in $\{0, \ldots, 2j - 1\}$ (appending $j - l$ further, strictly smaller, distinct exponents) exactly when $b_{l-1} \geq j - l$: append $b_{l-1} - 1, \ldots, b_{l-1} - (j - l)$ when this holds, and no extension exists when it does not, since fewer than $j - l$ nonnegative integers lie below b_{l-1}. Writing $c_i := b_i - (j - l)$ turns $b_{l-1} \geq j - l$ into $c_{l-1} \geq 0$ and $b_0 \leq 2j - 1$ into $c_0 \leq 2j - 1 - (j - l) = j + l - 1$, and $\sum_i 2^{b_i} 3^i = 2^{j-l} \sum_i 2^{c_i} 3^i$. So the top-$l$-exponent tuples realizable by $R_{j-1,j}$ are exactly 2^{j-l} times the decreasing l-tuples counted by $U(l, j+l-1)$, giving the claimed identity modulo 3^l. □

Proposition 15 (Existence)**.** *$j^*(l)$ exists for every $l \geq 1$, with $j^*(l) \leq 2 \cdot 3^{l-1} - 1$.*

Proof. $U(l, \cdot)$ is monotone increasing in W: enlarging the range $\{0, \ldots, W\}$ only adds admissible exponent tuples. Shifting every exponent of a tuple realizing $u \in U(l, W)$ up by 1 gives a decreasing l-tuple with top exponent at most $W + 1$ and bottom exponent at least $1 > 0$, still admissible, realizing $2u$; hence $2\,U(l, W) \subseteq U(l, W + 1)$ for every $W \geq l - 1$. The tuple $\beta_i = l - 1 - i$ $(i = 0, \ldots, l - 1)$ is admissible at $W = l - 1$ and gives $u_0 := \sum_i 2^{l-1-i} 3^i \equiv 2^{l-1} \pmod 3$, a unit, so $u_0 \in U(l, l - 1)$. Iterating the inclusion, $2^k u_0 \in U(l, l - 1 + k)$ for every $k \geq 0$. Since 2 has order $2 \cdot 3^{l-1}$ modulo 3^l (already used in Section 4), $\{2^k u_0 : 0 \leq k < 2 \cdot 3^{l-1}\}$ is the coset $u_0\langle 2\rangle = (\mathbb{Z}/3^l\mathbb{Z})^*$: every unit modulo 3^l appears as $2^k u_0$ for some k in that range, hence lies in

$U(l, l-1+k) \subseteq U(l, W^\dagger)$, $W^\dagger := l + 2\cdot 3^{l-1} - 2$, by monotonicity. So $U(l, W^\dagger) = (\mathbb{Z}/3^l\mathbb{Z})^*$, and Lemma 14 at $j = W^\dagger - l + 1 = 2\cdot 3^{l-1} - 1 \geq l$ gives $R_{j-1,j} \equiv (\mathbb{Z}/3^l\mathbb{Z})^* \pmod{3^l}$, hence $j^*(l) \leq 2\cdot 3^{l-1} - 1$. □

This bound is exponential, far above the conjectured $\log_4 3 \cdot l$ rate; it establishes only that $j^*(l)$ is a well-defined finite quantity, which the rest of this section and Section 4's mean-payoff-game argument then bound far more tightly.

By Lemma 14, which itself needs $j \geq l$, the smallest $j \geq l$ with $U(l, j+l-1) = (\mathbb{Z}/3^l\mathbb{Z})^*$ is the smallest $j \geq l$ at which $R_{j-1,j}$ covers; since $j^*(l) \geq l$ at every computed level (Table 1), with equality only at $l = 1$, this coincides with $j^*(l)$ itself throughout the range this paper works in, though nothing here rules out coverage at some unchecked $j < l$ for a level not yet computed. Writing $W^*(l) := j^*(l) + l - 2$ for the width one step before that,

$$H(l,j) = 2^{j-l}\, D(l, j+l-1), \qquad D(l,W) := (\mathbb{Z}/3^l\mathbb{Z})^* \setminus U(l,W). \tag{5}$$

Theorem 16 (Last-holdout parity)**.** *If $j^*(l) \geq l+1$, every element of $H(l, j^*(l)-1)$ is $\equiv 1 \pmod 3$. This covers $l = 2, \ldots, 23$ (Table 1); the hypothesis is inapplicable only at $l = 1$ ($j^*(1) = 1$, outside (5)'s range), which says nothing about whether the conclusion itself would hold there, and would become inapplicable again for any l at which the conjectured rate $\log_4 3 < 1$ eventually pushes $j^*(l)$ below $l+1$.*

Proof. $U(l,\cdot)$ is monotone increasing in W, so $D(l,\cdot)$ is monotone decreasing; every $x \in D(l, W^*(l))$ is, by minimality of $j^*(l)$, representable at width $W^*(l)+1$ but not at $W^*(l)$. Any witness for x at width $W^*(l)+1$ using only exponents $\leq W^*(l)$ would already witness x at that smaller width, so every such witness has top exponent $\beta_0 = W^*(l)+1$. Its contribution to the value, at rank 0, is $2^{\beta_0}\cdot 3^0 = 2^{W^*(l)+1}$, so $x \equiv 2^{W^*(l)+1} \equiv (-1)^{W^*(l)+1} \pmod 3$. By (5), which needs $j^*(l) - 1 \geq l$, elements of $H(l, j^*(l)-1)$ are $2^{j^*(l)-1-l}x$ for such x, hence

$$2^{j^*(l)-1-l}x \equiv (-1)^{j^*(l)-1-l}\cdot(-1)^{W^*(l)+1} = (-1)^{2j^*(l)-2} = 1 \pmod 3,$$

using $W^*(l)+1 = j^*(l)+l-1$ and that $2j^*(l)-2$ is always even. □

Empirical Result 17 (Near-extinction bijection)*.* $H(l, j^*(l)-1) = 2\cdot\{x \in H(l, j^*(l)-2) : x \equiv 2 \pmod 3\}$, exactly, verified at every $l = 2, \ldots, 9$ against the exact table with no exception.

The same width mechanism that proves Theorem 16 governs this bijection one step earlier, through the second-highest exponent β_1 of the extremal witness; unlike the top-exponent argument above, pinning β_1 down needs the one-step width identity of Lemma 21 below. We have not carried that argument through to a full proof of this bijection and record it as empirically verified only.

A natural cross-level congruence relating $H(l+1, j^*(l+1)-1)$ to $H(l, j^*(l)-1)$ does not hold. Writing π_l for reduction modulo 3^l, the candidate congruence $x \equiv 4\,\pi_l(x) \pmod{3^{l+1}}$ is impossible whenever $\pi_l(x)$ is a unit and $l \geq 2$: reducing both sides modulo 3^l forces $\pi_l(x) \equiv 4\,\pi_l(x) \pmod{3^l}$, i.e. $3\,\pi_l(x) \equiv 0 \pmod{3^l}$. Checked exhaustively at $l = 2, \ldots, 6$, $\pi_l(x)$ never lies in $H(l, j^*(l)-1)$ for $x \in H(l+1, j^*(l+1)-1)$ in the first place, so the pattern has no witnessing instance at these levels either. No cross-level relation between $H(l+1,\cdot)$ and $H(l,\cdot)$ is established by this paper.

Proposition 18 (Mod-9 two-class containment)**.** *Writing $J := j^*(l)$, if $J \geq l+1$ then $H(l, J-1) \bmod 9 \subseteq \{4^J, 4^{J+1}\} \pmod 9$.*

Proof. As in Theorem 16's proof, every witness of $x \in D(l, W^*(l))$ at width $W^*(l)+1$ has top exponent $\beta_0 = W^*(l)+1$; modulo 9, only the rank-0 and rank-1 terms survive (rank $i \geq 2$ contributes a multiple of 3^2), so $x \equiv 2^{\beta_0} + 3\cdot 2^{\beta_1} \pmod 9$ for the witness's second-highest exponent β_1. By (5), elements of $H(l, J-1)$ are $y = 2^{J-1-l}x$, so $y \equiv 2^{J-1-l+\beta_0} + 3\cdot 2^{J-1-l+\beta_1} \pmod 9$; using $\beta_0 = W^*(l)+1 = J+l-1$, the first term is $2^{2J-2} = 4^{J-1} \bmod 9$, fixed. The second term, $3\cdot 2^m \pmod 9$ for $m := J-1-l+\beta_1$, depends only on the parity of m (since $2^m \bmod 3$ has period

2), taking exactly the two values 3 or 6 $\pmod 9$ regardless of β_1's actual value; a direct check gives $\{4^{J-1}+3, 4^{J-1}+6\} \equiv \{4^J, 4^{J+1}\} \pmod 9$ for every J. So $y \bmod 9 \in \{4^J, 4^{J+1}\}$. ☐

Empirical Result 19 (Exclusion of the lower class). The class actually occupied is 4^{J+1}, never 4^J, at every level checked, $l = 3, \ldots, 16$ (extending the previous computational range, $l = 3, \ldots, 9$). Excluding 4^J in general is the sharpest concrete open question this part of the investigation produced: Proposition 18 places no constraint on β_1 itself, only on the parity-dependent outcome, so ruling out one of the two parities needs a different argument.

Proposition 20 (Sufficiency of bounded maxrun)**.** *If* $\operatorname{maxrun}(H(l, l+1))$ *is bounded by a constant* C *for all* l*, then* $j^*(l) \le l + C + 1$*.*

Proof. Corollary 11 at $j = l + 1$. ☐

The converse is not established, and the natural argument for it does not work: reading Corollary 11 downward gives only $\operatorname{maxrun}(H(l, l+1)) \ge j^*(l) - l - 1$, a *lower* bound on maxrun, which does not force maxrun to stay bounded when $j^*(l) - l$ does. Proving the converse would need an upper bound on maxrun in terms of $j^*(l) - l$, equivalent to a tightness statement for Corollary 11 that Section 8's corner-redundancy question addresses and that is false in the strong, per-witness form: recall 1547 from Section 6.

A naive expectation, that a random set of holdouts at the observed density would have doubling chains far longer than the flat ceiling of 3–4 measured at every level $l = 5, \ldots, 22$, does not survive contact with the actual, rapidly declining density of $H(l, l+1)$. An independence model matched to that actual density tracks the observed sequence within about one unit at every level, including both the one rise and the one fall in it: $\operatorname{maxrun}(H(l, l+1))$ is typical for the set's density under that model. This does not, by itself, say anything about whether $\operatorname{maxrun}(H(l, l+1))$ is bounded; Proposition 20 is the only established link between that question and $j^*(l)$'s rate.

Lemma 21 (One-step width identity)**.** *Let* $\operatorname{Corner}(l, W+1)$ *be the set of values realized by a witness tuple* $\beta_0 > \cdots > \beta_{l-1}$ *with* $\beta_0 = W + 1$ *and* $\beta_{l-1} = 0$ *simultaneously. Then*

$$U(l, W+1) = U(l, W) \cup 2U(l, W) \cup \operatorname{Corner}(l, W+1).$$

Proof. $\supseteq$: $U(l, W) \subseteq U(l, W+1)$ by monotonicity, $2U(l, W) \subseteq U(l, W+1)$ by the shift argument of Proposition 15's proof, and $\operatorname{Corner}(l, W+1) \subseteq U(l, W+1)$ since a corner witness is, in particular, a witness at width $W + 1$.

$\subseteq$: let $u \in U(l, W+1)$, witnessed by $W + 1 \ge \beta_0 > \cdots > \beta_{l-1} \ge 0$. If $\beta_0 \le W$, the tuple already witnesses u at width W, so $u \in U(l, W)$. If $\beta_0 = W + 1$ and $\beta_{l-1} \ge 1$, subtracting 1 from every exponent gives a decreasing tuple in $[0, W]$ (top exponent W, bottom exponent ≥ 0) witnessing $u/2$, so $u \in 2U(l, W)$. If $\beta_0 = W + 1$ and $\beta_{l-1} = 0$, this is exactly the corner case, so $u \in \operatorname{Corner}(l, W+1)$. These three cases are exhaustive. ☐

At every width W in the range $2l + 1 \le W \le j^*(l) + l - 1$, checked exhaustively for $l = 3, \ldots, 13$ (the upper end is the largest width the exact table reaches at each level), $\operatorname{Corner}(l, W+1)$ contributes nothing new: every corner witness's value is already covered without it. Call this the *corner-redundancy* property at width W; whether it holds at every width $W \ge 2l + 1$ in general is unproven, with the only known failures at the boundary width $W = 2l$, which is exactly $j = l + 1$, Proposition 20's own case.

Proposition 22 (Corner-redundancy implies tightness)**.** *Fix* j *with* $l + 2 \le j \le j^*(l)$*. If corner-redundancy holds at every width* $2l + 1 \le W \le j^*(l) + l - 1$*, then* $j^*(l) = j + \operatorname{maxrun}(H(l, j))$ *exactly.*

Proof. Corner-redundancy at width W says $\operatorname{Corner}(l, W+1) \subseteq U(l, W) \cup 2U(l, W)$, so Lemma 21 gives $U(l, W+1) = U(l, W) \cup 2U(l, W)$. Multiplication by 2 is a bijection of $(\mathbb{Z}/3^l\mathbb{Z})^*$, so it commutes with complementation and with intersection; taking complements turns this into $D(l, W + 1) = D(l, W) \cap 2D(l, W)$, and pushing through the shift in (5) (itself a bijection, for the same reason) turns the width-level equality at $W = j' + l - 1$ into an equality of holdout sets, $H(l, j' + 1) = 2H(l, j') \cap 4H(l, j')$, sharpening Theorem 9's inclusion to an identity whenever corner-redundancy holds at that width.

Claim: for every j' with $j \leq j' < j^*(l)$, $\mathrm{maxrun}(H(l, j' + 1)) = \mathrm{maxrun}(H(l, j')) - 1$, given corner-redundancy at width $j' + l - 1$. The inequality $\leq$ is unconditional: a maximal chain of length $\mathrm{maxrun}(H(l, j'+1))$ in $H(l, j'+1)$ extends, by Corollary 10, to a chain one longer in $H(l, j')$. For $\geq$, since $j' < j^*(l)$, minimality of $j^*(l)$ gives $H(l, j') \neq \emptyset$, so $s := \mathrm{maxrun}(H(l, j')) \geq 1$; fix a maximal chain $x, 2x, \ldots, 2^{s-1}x \in H(l, j')$. For each $k = 2, \ldots, s$, both $2^{k-1}x$ and $2^{k-2}x$ lie in this chain, so the equality $H(l, j' + 1) = 2H(l, j') \cap 4H(l, j')$ places $2^k x \in H(l, j' + 1)$; this gives the $s - 1$ consecutive powers $4x, 8x, \ldots, 2^s x \in H(l, j' + 1)$, so $\mathrm{maxrun}(H(l, j' + 1)) \geq s - 1$, proving the claim.

The widths this uses, $j'+l-1$ for $j \leq j' < j^*(l)$, all lie in $[j+l-1, j^*(l)+l-2] \subseteq [2l+1, j^*(l)+l-1]$ (using $j \geq l + 2$), inside the hypothesis, regardless of $\mathrm{maxrun}(H(l, j))$'s actual value. Telescoping the claim over every step from $j' = j$ to $j' = j^*(l) - 1$ (a fixed number of steps, $j^*(l) - j$, fixed before any of this) gives $\mathrm{maxrun}(H(l, j^*(l))) = \mathrm{maxrun}(H(l, j)) - (j^*(l) - j)$. The left side is 0 ($H(l, j^*(l)) = \emptyset$ by definition), so $\mathrm{maxrun}(H(l, j)) = j^*(l) - j$, i.e. $j^*(l) = j + \mathrm{maxrun}(H(l, j))$. $\square$

Since corner-redundancy is verified at every width the proof above uses for $l = 3, \ldots, 13$, Proposition 22 is unconditional at those levels: Empirical Result 12's equality is a proven theorem there, for every j with $l + 2 \leq j \leq j^*(l)$. In general, whether this implication's hypothesis holds is exactly the corner-redundancy question above: it does not, by itself, bear on Proposition 20's converse, which turns on the single boundary budget $j = l + 1$ ($W = 2l$), exactly where corner-redundancy is known to fail.

8. Discussion

Two named questions summarize what stands between this paper's results and a sharper theorem. Excluding the lower class in Proposition 18's containment (Empirical Result 19) would turn the observed single-class pattern into a fully proven statement. Proving corner-redundancy at every width $W \geq 2l + 1$ would, by Proposition 22, make Corollary 11's bootstrap tight for every budget $j \geq l + 2$ in general, not merely at the levels $l = 3, \ldots, 13$ where corner-redundancy is already verified; it would not, by itself, settle Proposition 20's converse, which turns on the single boundary budget $j = l + 1$ ($W = 2l$), exactly where corner-redundancy is known to fail. Neither question is, on its face, as hard as the Weak Covering Conjecture itself; whether either is substantially easier is not known.

A finite exponential sum built from strictly decreasing, exponentially weighted digit tuples is a natural object beyond this specific covering problem: finite-population L-statistics at non-CLT frequencies and weights, Littlewood–Offord theory for structured coefficient sequences, and the analytic side of the Collatz literature via Tao's discussion are all adjacent areas. Three things found here may recur there under different names: generic cancellation unable to match a bound already conditionally established here by other means, exceptional mass that does not concentrate on a sparse set at the accessible scale, and coverage failure decided by an extremal arithmetic obstruction that a phase-randomization experiment showed magnitude-only methods miss. Tao's later, unrelated result that almost every Collatz orbit attains an almost bounded value [3] works with a first-passage random variable over a different probabilistic model of the same map and does not bear on covering directly, but is the clearest instance of a new kind of technique changing what counts as tractable in this area, the kind of shift a resolution of either open question above would need. Predecessor-density bounds on the $3n + 1$ orbit structure itself, most developed in Krasikov and Lagarias's difference-inequality approach [4] (a lower bound of $x^{0.84}$ on the count of integers below x whose orbit reaches 1), target the same broad question as Wirsching's own reduction from covering to positive predecessor density, by a different mechanism: orbit-level difference inequalities rather than the covering structure of one finite family $R_{j-1,j}$. Nothing in that literature currently transfers to $K(l)$ or $j^*(l)$ directly, but the shared target makes a transfer more plausible than an unrelated result would, and whether one exists is itself an open question.

The Weak Covering Conjecture itself, and the exact asymptotic rate of $e(l)$, remain open.

9. Code and data availability

The repository https://github.com/faculdade/weak-covering-conjecture holds the code and data behind every computation reported here, one folder per section. It contains the covering search that produced Table 1, the model comparison behind Table 2, the mean-payoff-game solver, the exponential-sum computations of Section 5, and the holdout computations of Sections 6 and 7, together with the stored holdout sets at the final budgets of each level $l = 5, \ldots, 21$. The twelve certificates (σ_k, ρ_k, h_k) behind Table 3 are stored there with a verifier that re-checks each of them against the inequality in Theorem 4's proof, without re-running the solver. Each folder's README gives what it checks, the command, the output to expect, and what the run costs. Most checks take seconds; the three largest levels of Table 1 do not.

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